Darshan UNIVERSITY

योग: कर्मसु कौशलम्

Python Programming - 2301CS404

Lab - 5

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1) Character Category Counter

Problem: Count uppercase, lowercase, digits, and special characters in a string.

Sample Inputs & Outputs:

- Input: `Python3@Lab` → Uppercase: 2, Lowercase: 7, Digits: 1, Special: 1
- Input: `HELLO123` → Uppercase: 5, Lowercase: 0, Digits: 3, Special: 0
- Input: `hi@#9` → Uppercase: 0, Lowercase: 2, Digits: 1, Special: 2

In [14]:
```
s = input("Enter String");
uc = 0;
lc = 0;
d = 0;
sc = 0;
for i in s:
    if i.islower():
        lc = lc+1;
    elif i.isupper():
        uc = uc+1;
    elif i.isdigit():
        d = d+1;
    else:
        sc = sc + 1;

print("Uppercase:",uc,"Lowercase:",lc,"Digit:",d,"Special:",sc);
```

```
Enter StringHELLO123
Uppercase: 5 Lowercase: 0 Digit: 3 Special: 0
```

2) Reverse Each Word

- Input: `Python is Easy` → `nohtyP si ysaE`
- Input: `Hello World` → `olleH dlroW`
- Input: `Learn Python Fast` → `nraeL nohtyP tsaF`

In [10]:
```
s = input("Enter String:")
rs = s.split(" ")
ans = ""
for i in rs:
    ans = ans+(i)[::-1]
    ans = ans + " "
print(ans)
```

```
Enter String:Hello World
olleH dlroW
```

3) Case Pattern Identifier

- `PYTHON` → Uppercase
- `python` → Lowercase
- `Python Programming` → Title Case

In [17]:
```
s = input("Enter String:")

if s.isupper():
    print("UpperCase")
elif s.islower():
    print("Lowercase")
elif s.title():
    print("Titlecase")
```

```
Enter String:python
Lowecase
```

4) First and Last Occurrence Finder

- `programming` , `g` → First:3 Last:10
- `banana` , `a` → First:1 Last:5
- `hello` , `z` → Not found

In [21]:
```
s = input("Enter String:")
ch = input("Enter to find Index:");

print(f"First:{s.index(ch)} Last:{s.rindex(ch)}")
```

```
Enter String:banana
Enter to find Indexa
First:1 Last:5
```

5) Word Frequency Counter (ignore the case)

- `Python is easy and Python is powerful` → python: 2
- `Java is popular but java is verbose` → java: 2
- `C is fast` → c:1

In [24]:
```
s = input("Enter String:")
ch = input("Enter word:")

s = s.lower()
ch = ch.lower()

print(f"{ch}:{s.count(ch)}")
```

```
Enter String:Java is popular but java is verbose
Enter word:java
java:2
```

6) Remove Extra Spaces

- `Python  is  fun` → `Python is fun`
- `Hello  World` → `Hello World`
- `NoExtraSpace` → `NoExtraSpace`

In [31]:
```
s = input("Enter String:")
ans = s.split()
print(" ".join(ans))
```

```
Python is fun
```

7) Prefix and Suffix Removal

- `unhappy.txt` → `happy`
- `pretest.py` → `test`
- `unwanted.doc` → `wanted`

In [38]:
```
s = input("Enter String:")
pr = input("Enter Prefix:")
sf = input("Enter Suffix:")

ans = s.replace(pr,"")
ans = ans.replace(sf,"")

print(ans)
```

```
Enter String:unhappy.txt
Enter Prefix:un
Enter Suffix:.txt
happy
```

8) Replace Vowels

- `Programming` → `Pr*gr*mm*ng`
- `Education` → `*d*c*t**n`
- `Sky` → `Sky`

In [37]:
```
s = input("Enter String:")

for i in s:
    if i.lower()=='a' or i.lower()=='e' or i.lower()=='i' or i.lower()=='o' or i.lc
        s = s.replace(i,'*')
print(s)
```

```
Enter String:Programming
Pr*gr*mm*ng
```

9) String Compression

- `aaabbcccc` → `a3b2c4`
- `xxxyyyz` → `x3y3z1`
- `abcd` → `a1b1c1d1`

In [63]:
```
s = input("Enter String:")
ans = ""
count = 1;
for i in range(1,len(s)):
    if s[i-1] == s[i]:
        count = count+1;
    else:
        ans = ans+ s[i-1]+str(count)
        count = 1;


ans = ans+ s[-1]+str(count)
print(ans)
```

```
Enter String:abcd
a1b1c1d1
```

10) Toggle Case (without using str.swapcase())

- `PyThOn` → `pYtHoN`
- `HELlo` → `helLO`
- `Python` → `pYTHON`

In [78]:
```
s = input("Enter String:")
ans = ""
for i in s:
    if i.islower():
        ans = ans + i.upper()
    else:
        ans = ans + i.lower()
print(ans)
```

```
Enter String:HELlo
helLO
```

11) Username Validation

valid username rules:

1. only alphabets and digits
2. does not start with a digit
3. length ≥ 6

- `User123` → Valid
- `123User` → Invalid
- `Us@12` → Invalid

In [47]:
```
s = input("Enter Username:")

if s.isalnum() and not s[0].isdigit() and len(s)>=6:
    print("Valid")
else:
    print("Invalid")
```

```
Enter Username:Us@12
Invalid
```

12) Palindrome Checker (ignore the case)

- `Madam` → Palindrome
- `Level` → Palindrome
- `Python` → Not Palindrome

In [51]:
```
s = input("Enter Sring:")
s = s.lower()
if s[0::] == s[-1::-1]:
    print("Palindrome")
else:
    print("Not Palindrome")
```

```
Enter Username:Python
Not Palindrome
```

13) Longest Word Finder

- `Python programming is interesting` → programming
- `I love coding` → coding
- `Data structures and algorithms` → structures

In [68]:
```
s = input("Enter String:")
max1 = 0;
l1 = s.split(" ")

ans = ""

for i in l1:
    if max(max1,len(i)) > max1:
        ans = i
        max1 = max(max1,len(i))
print(ans)
```

```
Enter String:Data structures and algorithms
structures
```

14) Case Conversion

- `python programming`
  - Upper: PYTHON PROGRAMMING
  - Lower: python programming
  - Title: Python Programming
  - Capitalize: Python programming

In [52]:
```
s = input("Enter String:")
```

```
print(f"Upper:{s.upper()} Lower:{s.lower()} Title:{s.title()} Capitalize:{s.capital
```

```
Enter Username:python programming
Upper:PYTHON PROGRAMMING Lower:python programming Title:Python Programming Capital
ize:Python programming
```

15) Custom Split

- `apple,banana,grape`
- `10|20|30`
- `A-B-C-D`

In [55]:

```
s = input("Enter String:")

sp = input("Enter for split:")

print(f"{s.split(sp)}")
```

```
Enter String:apple,banana,grape
Enter for split:,
['apple', 'banana', 'grape']
```

In [ ]:

```

```